



Breast Cancer Image Classification

Harnessing Deep Learning for Early and Accurate Diagnosis

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 [Project Repository](#)

Problem Statement

The Challenge of Diagnosis

Breast cancer remains a leading cause of mortality worldwide. Early detection is critical, yet manual histopathology analysis is:

-  **Time Intensive:** Requires hours of expert review.
-  **Error-Prone:** Subject to human fatigue and variability.
-  **Resource Heavy:** High demand for specialized pathologists.

Our Objective: Develop a robust binary classification system (Cancer vs. Non-Cancer) using state-of-the-art CNNs.



"Artificial intelligence can analyze tissue samples with precision that complements human expertise."

Dataset & Preprocessing

Data Acquisition

We utilized a histopathology dataset focused on binary detection of cancerous cells. To ensure model compatibility, we implemented a rigorous pipeline:

-  **Uniform Resizing:** All images standardized to 224x224 pixels.
-  **Normalization:** Pixel values rescaled by $1/255$ for faster convergence.
-  **Tech Stack:** NumPy, Pandas, and TensorFlow/Keras for handling high-volume data.

 Breast cancer histopathology slide microscopic image

Fig 1: Microscopic cell tissue samples used for training.

Model 1: Scratch CNN



Custom Architecture

We designed a Convolutional Neural Network from the ground up to learn features specific to our dataset without external bias.

- **Feature Extraction:** Successive layers of convolutions and max-pooling.
- **Activation:** ReLU for hidden layers to manage vanishing gradients.
- **Optimization:** Tuned specifically for binary cross-entropy loss.

Result: Achieved an impressive 96.4% accuracy, proving that a task-specific design can be highly effective.

Transfer Learning Models



ResNet50

Utilizes residual connections (skip connections) to train very deep networks effectively by solving the degradation problem.

F1-Score: 0.7619



DenseNet169

Connects each layer to every other layer in a feed-forward fashion, maximizing feature reuse and reducing parameters.

F1-Score: 0.8888

By using pre-trained weights from ImageNet, these models benefit from sophisticated low-level feature detection learned from millions of images.

Evaluation Metrics

-  **Accuracy:** The overall percentage of correct predictions (Cancer vs. Non-Cancer).
-  **Precision:** Reliability of positive predictions (minimizing false positives).
-  **Recall (Sensitivity):** Ability to find all actual cancer cases (minimizing false negatives).
-  **F1-Score:** The harmonic mean of Precision and Recall, providing a balanced measure.

Libraries used: **Scikit-learn** for metrics, **Matplotlib** & **Seaborn** for visualization.

Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Scratch CNN	0.9643	1.0000	0.7895	0.8824
ResNet50	0.9107	0.6957	0.8421	0.7619
DenseNet169	0.9643	0.9412	0.8421	0.8889



Conclusion

Our study confirms that deep learning models can effectively distinguish between cancerous and non-cancerous tissue slides with over 96% accuracy.

Future Directions:

-  **Multi-class Classification:** Grading tumor severity levels.
-  **Clinical Validation:** Testing on real-world unseen hospital data.
-  **Deployment:** Creating a web/mobile interface for pathologist assistance.

Thank You! Any Questions?

Image Sources



Thumbnail
for

https://www.mdpi.com/diagnostics/diagnostics-15-00582/article_deploy/html/images/diagnostics-15-00582-g003a-550.jpg

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